

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIM**

H2 BIOLOGY**9744/2****Paper 2**

Monday

10th September 2018

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' Use	
Section A	
1	/13
2	/13
3	/14
4	/14
5	/16
6	/10
7	/9
8	/5
9	/6
Total	/100

This document consists of **22** printed pages.

[Turn over

QUESTION 1

Fig 1.1 shows the structure of a G-protein-linked receptor (GPLR) from a cross-section of the plasma membrane.

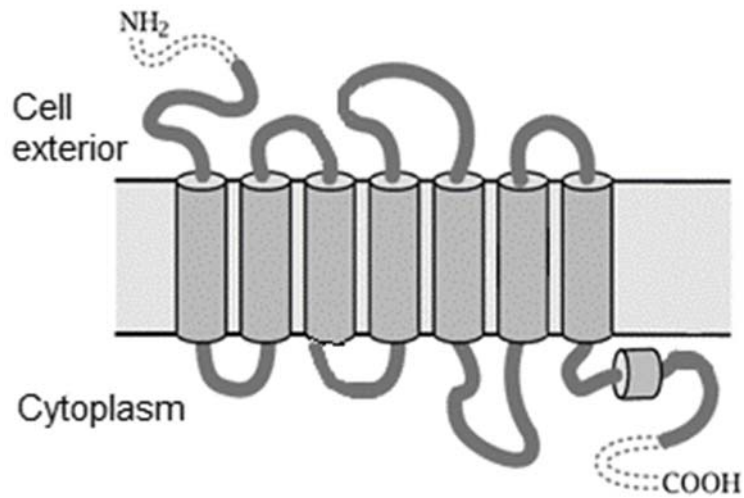


Fig 1.1

(a) With reference to **Fig 1.1**, describe the significance of R groups to the structure and function of a GPLR.

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- (b) The GPLRs make up the largest family of cell surface receptors. Outline the route taken by the GPLR after its synthesis to its final location in the plasma membrane.

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GPLRs are found to be closely associated with a type of G protein called K-Ras in the cell signaling pathways.

Fig. 1.2 is a simplified diagram showing the normal roles of GPLR and K-Ras in the RAS/MARK signaling pathway.

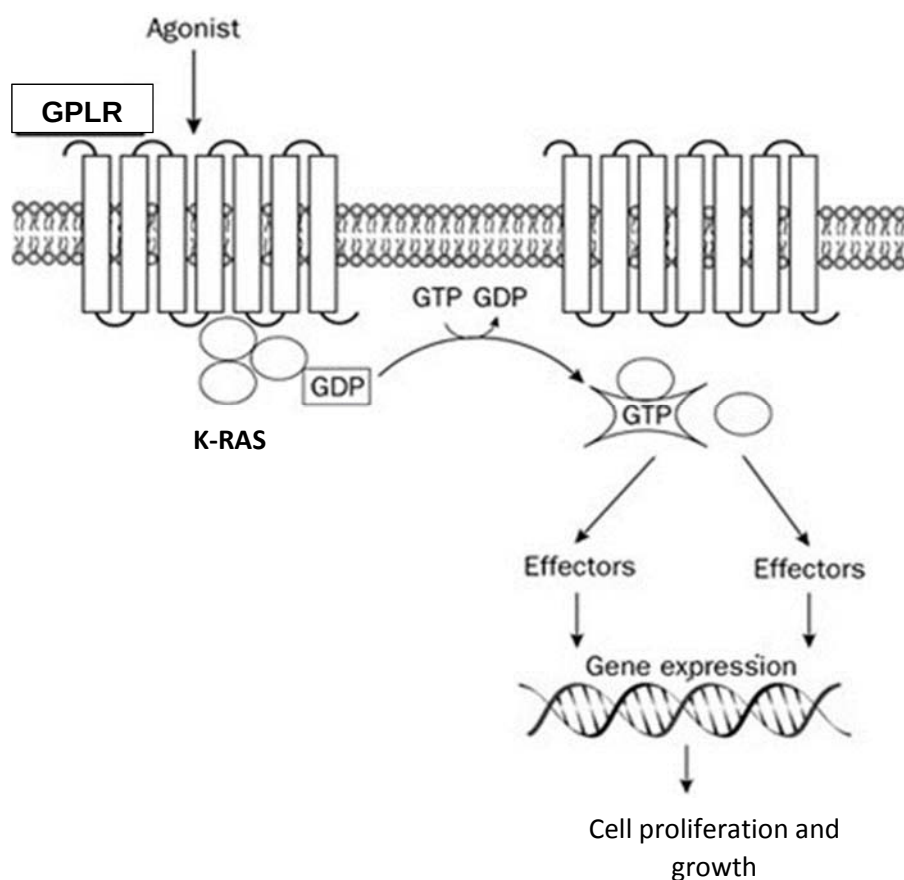


Fig. 1.2

(c) Assuming that the effectors (in **Fig. 1.2**) in the transduction pathways function normally, explain how a mutation can lead to the formation of tumours in cancer.

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[Total: 13m]

QUESTION 2

Fig. 2.1 shows Process X in an eukaryotic cell which produces ribosomal RNA (rRNA).

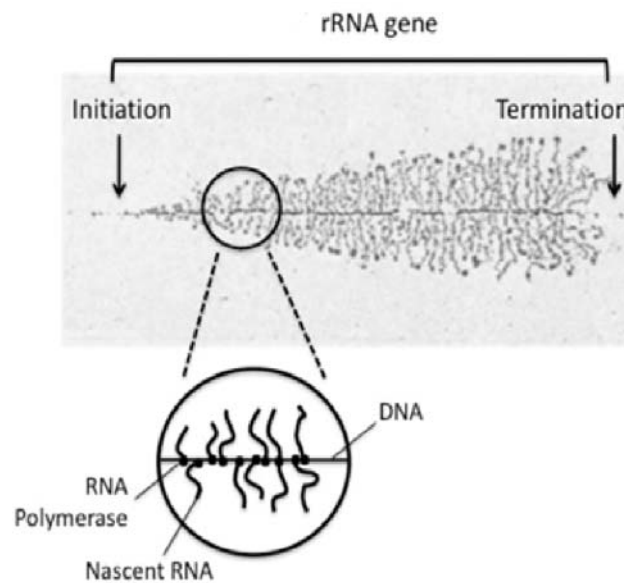


Fig. 2.1

(a)(i) Name the Process X occurring in Fig. 2.1.

.....[1]

(ii) List one molecule **not mentioned in Fig. 2.1** that is required for Process X.

.....[1]

(iii) Describe how RNA polymerase is able to recognise and bind to the promoter on DNA and not to other DNA regions.

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 [2]

(iv) Explain for the observed pattern of Process X in **Fig. 2.1**.

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..... [2]

(v) State the roles of rRNA in protein synthesis.

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..... [2]

(b) During protein synthesis in cells of an embryo, all tRNA molecules with UAC anticodon sequence, are observed to be bound to arginine amino acid instead of methionine.

(i) Suggest how these tRNA molecules attached with the wrong amino acid might arise.

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..... [2]

(ii) Suggest and explain the effect of this wrong pairing of amino acid to tRNA on the embryo.

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..... [3]

[Total: 13m]

QUESTION 3

In a dihybrid inheritance, gene B/b codes for flower colour while gene H/h codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then undergo a test cross.

(a) Predict the expected phenotypic ratio in the F2 progeny.

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..... [1]

(b) Explain how different characteristics can be inherited independently in dihybrid inheritance.

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..... [2]

(c) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross if inheritance is Mendelian.

.....[3]

(d) You will now proceed to do a chi-square statistical test for the above cross.

(i) State the objective of performing a chi-squared statistical test.

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 [1]

The number and phenotypes of F₂ plants are listed below:

F₂ phenotypic classes	Observed numbers
Red flower, fan-shaped leaf	85
Red flower, oval leaf	70
Yellow flower, fan-shaped leaf	89
Yellow flower, oval leaf	78

Formula for χ^2 calculation

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'
 ν = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

(ii) Calculate the chi-square value.

.....[2]

Table 3.1

degrees of freedom	probability P				
	0.50	0.10	0.05	0.01	0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.35	16.27
4	3.36	7.78	9.49	13.28	18.47

(iii) Using Table 3.1 and the calculated chi-square value, find the probability that observed and expected results differ by chance.

.....[1]

(iv) State the conclusions for this test.

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..... [2]

(v) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

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..... [2]

[Total: 14m]

QUESTION 4

(a) In many ant species, polymorphism exists in the form of worker and queen ants. Some distinct differences between workers and queens are that workers are much smaller and cannot reproduce. Strict caste roles are also observed: queens lay eggs and workers take care of all other work, including offspring.

In a new study to identify the cause for these behavioural and physical differences, Rockefeller scientists report that a gene coding for an insulin-like peptide, ILP2, is instrumental in promoting and suppressing reproduction.

ILP2 is the ant version of insulin and, like human insulin, regulates metabolism by cellular uptake of glucose.

The table below summarises their results:

Expression of ILP2	Type of ants
High	Reproducing
Low	Non-reproducing

(i) Suggest a link between glucose uptake and reproduction.

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 [1]

(ii) The presence of larvae causes the activation of ovaries in worker ants. It was suggested that larvae release pheromones that control the expression of the ILP2 gene.

Explain how pheromones could have played a role at the transcriptional level of ILP2 expression in determining the caste roles of ants.

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 [4]

- (iii) The expression of the ILP2 gene can be further controlled even after translation of the ILP2 mRNA. Describe 2 ways to control ILP2 expression at this level.

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 [2]

- (b) Double-stranded DNA of the ILP2 gene is denatured and allowed to hybridise through complementary base pairing with mature ILP2 mRNA isolated from queen ant cells. This is shown in **Fig. 4.1**.

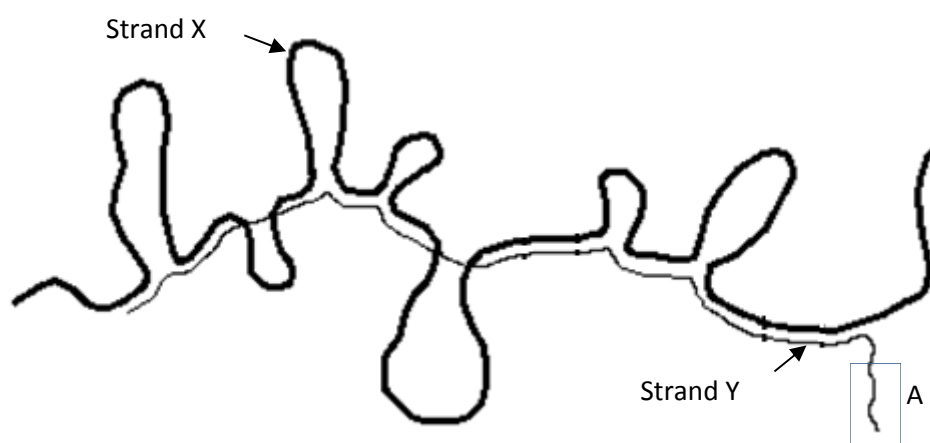


Fig 4.1

- (i) State the identity of Strand X.

.....[1]

- (ii) Explain your answer in (i) using evidence from **Fig. 4.1**.

.....

 [2]

- (iii) Circle the locations of the promoter and termination regions of the ILP2 gene on **Fig. 4.1**. Label the regions accordingly. [2]

- (iv) Suggest the identity of the unpaired segment (labelled A) on strand Y. Explain your answer.

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 [2]

[Total: 14m]

QUESTION 5

(a) The following extract was summarised from an article published in Proceedings of the National Academy of Sciences journal.

The article proposes a method to overcome HIV pandemic using genetically modified rice.

Summary points of article:

- Scientists from the US, UK, and Spain have developed a new strain of genetically modified rice to manage HIV symptoms in countries where traditional medicines can be hard to access.
- The rice seeds produce three proteins – the antibody 2G12, and the lectins griffithsin and cyanovirin-N – which preliminary *in vitro* tests show bind to gp120 and *neutralize* HIV.
- These seeds can be ground up to form a paste that can then be applied as a topical cream, which counterbalances the virus in the exact same way as the anti-retroviral medication.
- There remains a few *hurdles* researchers will have to jump before the rice becomes widely available.

(i) Define the term “neutralise” in this context and explain how this process can halt the reproductive life cycle of HIV.

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..... [3]

(ii) Suggest one possible hurdle that researchers need to overcome for this treatment using GM rice.

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..... [1]

(b) Fig 5.1 shows a lentivirus, which can bind to cells lining the airways of the lungs. The lentivirus is a form of **retrovirus**. The **general structure** of this virus is similar to that of HIV.

The lentivirus is commonly used as a vector (vehicle to transport external copies of RNA coding for specific proteins into cells).

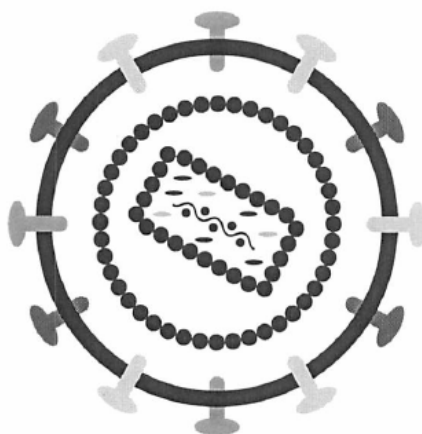


Fig. 5.1

(i) Using the letter **R**, label **Fig 5.1** to identify a feature that is protein in nature.
[1]

(ii) Explain why external copies of RNA intended to be introduced into cells cannot pass through the membrane of cells directly.

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 [2]

(iii) With reference to your knowledge on retroviruses, explain how **long-term** expression of an inserted RNA is brought about following infection of host cells with the lentiviral vector.

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 [3]

(c) Plasmids are small, self-replicating circles of DNA.

During cloning, a target gene can be inserted into a plasmid, making the plasmid a **vector** that transports the target gene into cells.

It is hoped that the plasmid and the target gene gets replicated to multiple copies in the cell.

Based on its features, suggest why plasmids are good choices of cloning vectors.

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.....[1]

(d) The F plasmid is a crucial plasmid found naturally in bacterial cells. It plays an integral role in the transfer of genes from one bacterial cell to another through processes such as conjugation.

Fig. 5.2 shows the process of conjugation.

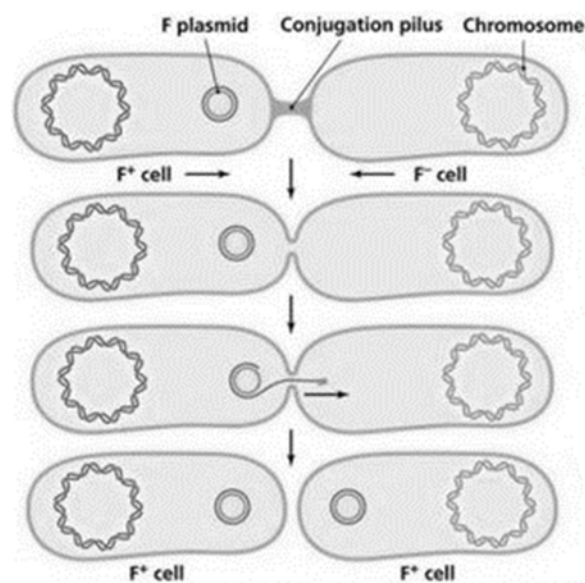


Fig. 5.2

During conjugation, the transferred F plasmid strand undergoes DNA replication similar to the synthesis of Okazaki fragments.

Explain why.

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..... [2]

(e) Scientists isolated one of the *lac permease* gene from the *lac* operon and proceeded to amplify the gene using an automated process of Polymerase Chain Reaction (PCR).

Fig. 5.3 shows the temperatures used at the different stages during PCR.

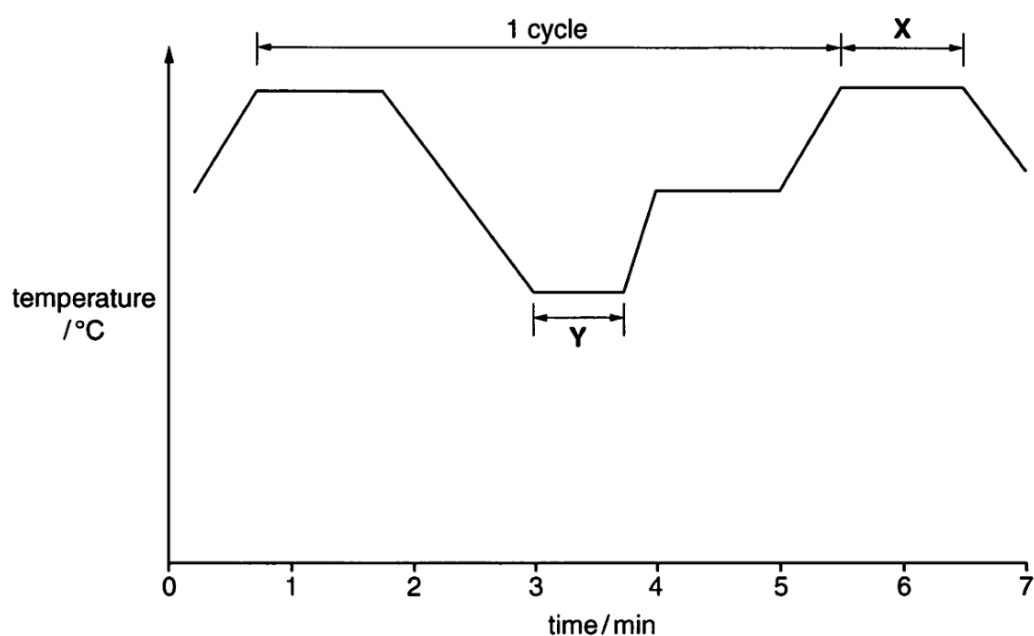


Fig. 5.3

(i) Describe what happens at stage Y.

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..... [2]

(ii) If stage X of PCR process persisted for more than two hours, suggest what you will expect to find in the PCR mixture during this period.

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..... [1]

[Total: 16m]

QUESTION 6

(a) The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million).

An experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

(i) Predict the concentration of carbon dioxide in the air after flowing over the leaves when the plant was placed in the dark.

..... [1]

(ii) Explain the experimental results obtained in relation to light intensity being a limiting factor of photosynthesis.

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 [3]

(b) Discuss the validity of the statement “The synthesis of ATP during photophosphorylation depends on the **transport of protons** across membranes.”

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 [4]

- (c) NADP⁺ is an important electron carrier found in plants, but its level decreases sharply in the day. Suggest the significance of the decrease of NADP⁺ during the day.

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..... [2]

[Total: 10m]

QUESTION 7

Ants and bees – which by all appearances seem so different – are found to be close relatives (refer to **Fig. 7.1**) in a new study which used cutting-edge DNA sequencing techniques to elucidate the taxonomic relationships among the different families of wasps, bees and ants.

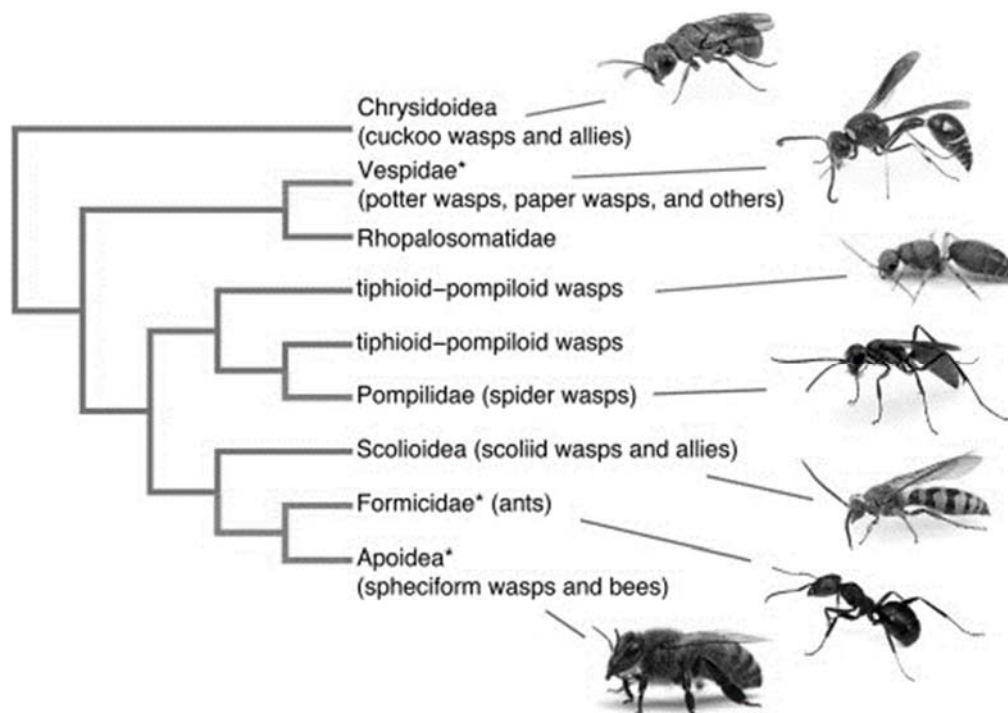


Fig. 7.1

- (a) State **one** advantage of using DNA sequences to elucidate the evolutionary relationship between ants and wasps.

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 [1]

- (b) Discuss how molecular homology was used in this study to derive the phylogram in **Fig. 7.1**.

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 [2]

(c) Explain how biogeography can help to support the evolutionary deductions in this case study.

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..... [2]

(d) The same study also suggested that when the early bees switched from preying on other insects to pollen feeding, the number of bee species exploded compared with their hunting wasp sister group. The switch from predatory behavior in hunting wasps to pollen feeding in bees also saw a corresponding increase in diversification of the mouthparts and tongue lengths of the bees.

Suggest and explain how the increase in number of bee species is made possible with the switch to pollen feeding, as compared to speciation in hunting wasps.

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[Total: 9m]

QUESTION 8

(a) Explain how the structure of antibodies cause pathogens to be destroyed by macrophages.

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.....[3]

(b) DNA coding for the light chain of an antibody is isolated from a **mature B** lymphocyte.

DNA coding for the light chain of an antibody is also isolated from a **developing B** lymphocyte (before maturation).

Gel electrophoresis was performed on both sources of DNA.

Fig 8.1 shows the gel diagram and the corresponding positive / negative electrode.

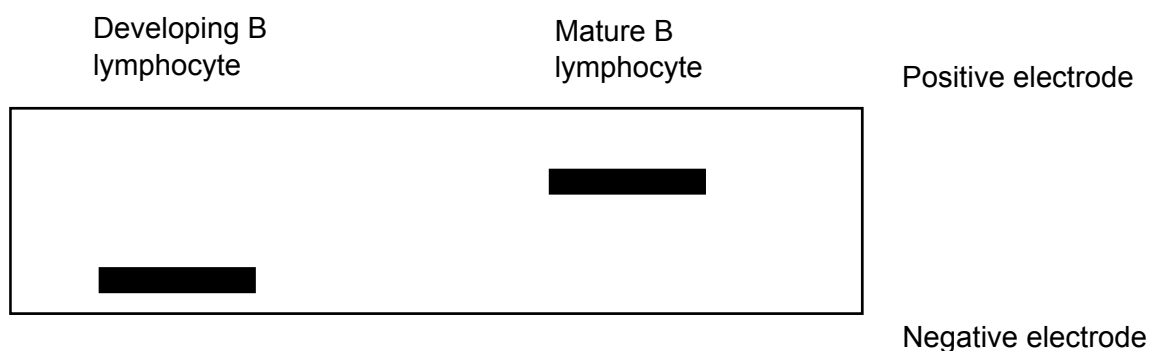


Fig. 8.1

Explain the difference in the results on the gel diagram.

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.....[2]

[Total: 5m]

QUESTION 9

Carbon dioxide is one of the greenhouse gases that contribute to global warming and subsequently climate change.

A group of high school students decided to test whether varying temperatures would correspondingly affect mean carbon dioxide gas emission.

Table 9.1 shows the mean carbon dioxide gas emission after exposing a fixed number of mealworms to different temperatures. Carbon dioxide gas emission was measured before and after exposure to experimental temperature.

Table 9.1 showing effects of varying temperatures on mean carbon dioxide gas emission, measured in parts per million (ppm). Values represent mean $\pm$ standard deviation.

Temperature /°C	Mean Carbon dioxide gas emission /ppm	
	Before exposure to experimental temperature	After exposure to experimental temperature
30.0	445 $\pm$ 25	450 $\pm$ 17
40.0	450 $\pm$ 20	500 $\pm$ 30
50.0	460 $\pm$ 17	540 $\pm$ 18

(a)(i) Describe the patterns shown by the data in Table 9.1.

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(ii) With reference to the mean carbon dioxide gas emission **before** exposure to 40 °C, suggest what standard deviation means.

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(b) Corals are affected by rising temperatures in ocean waters. Explain how.

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[Total: 6m]